Networking Lab – Multiple PCs Connected via Hub

Objective

Establish connectivity between multiple PCs (minimum 3, here 4 PCs: A, B, C, D) using one Hub and copper straight-through cables.

Topology Setup

- Devices:
  - 1 Hub
  - 4 PCs (A, B, C, D)
- Cable Type:
  - Copper Straight-Through
- Connections:
  - Each PC directly connected to the Hub

IP Address Configuration (Class C Network)

Network: 192.168.1.0/24
Class: Class C

Device	IP Address	Subnet Mask	Class
PC A	192.168.1.1	255.255.255.0	Class C
PC B	192.168.1.2	255.255.255.0	Class C
PC C	192.168.1.4	255.255.255.0	Class C
PC D	192.168.1.3	255.255.255.0	Class C

Cisco Packet Tracer - C:\Users\sms\Desktop\SMS\SMS_ 2024BCS0055_Lab4.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 1798, y: 217 | Root | 02:27:30

A

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration
- DHCP / Static
- IPv4 Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

IPv6 Configuration
- Automatic / Static
- IPv6 Address:
- Link Local Address: FE80::290:21FF:FE7D:36C3
- Default Gateway:
- DNS Server:

802.1X
- Use 802.1X Security
- Authentication: MD5
- Username:

Top

B

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration
- DHCP / Static
- IPv4 Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

IPv6 Configuration
- Automatic / Static
- IPv6 Address:
- Link Local Address: FE80::2D0:58FF:FE89:EB8D
- Default Gateway:
- DNS Server:

802.1X
- Use 802.1X Security
- Authentication: MD5
- Username:
- Password:

Top

C

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration
- DHCP / Static
- IPv4 Address: 192.168.1.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

IPv6 Configuration
- Automatic / Static
- IPv6 Address:
- Link Local Address: FE80::260:3EFF:FEA2:6281
- Default Gateway:
- DNS Server:

802.1X
- Use 802.1X Security
- Authentication: MD5
- Username:

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D

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration
- DHCP / Static
- IPv4 Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

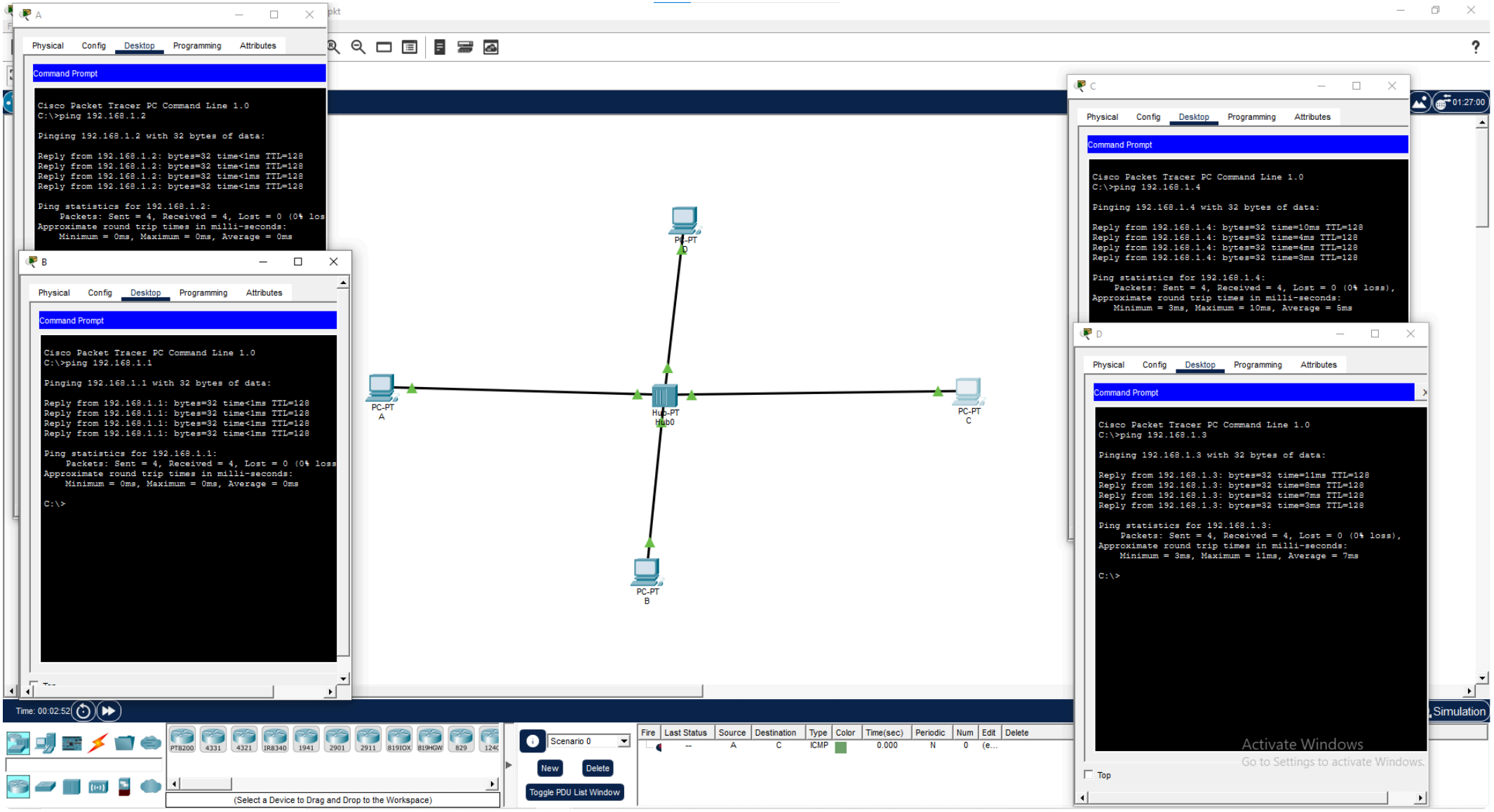
IPv6 Configuration
- Automatic / Static
- IPv6 Address:
- Link Local Address: FE80::260:2FFF:FE78:146E
- Default Gateway:
- DNS Server:

802.1X
- Use 802.1X Security
- Authentication: MD5
- Username:

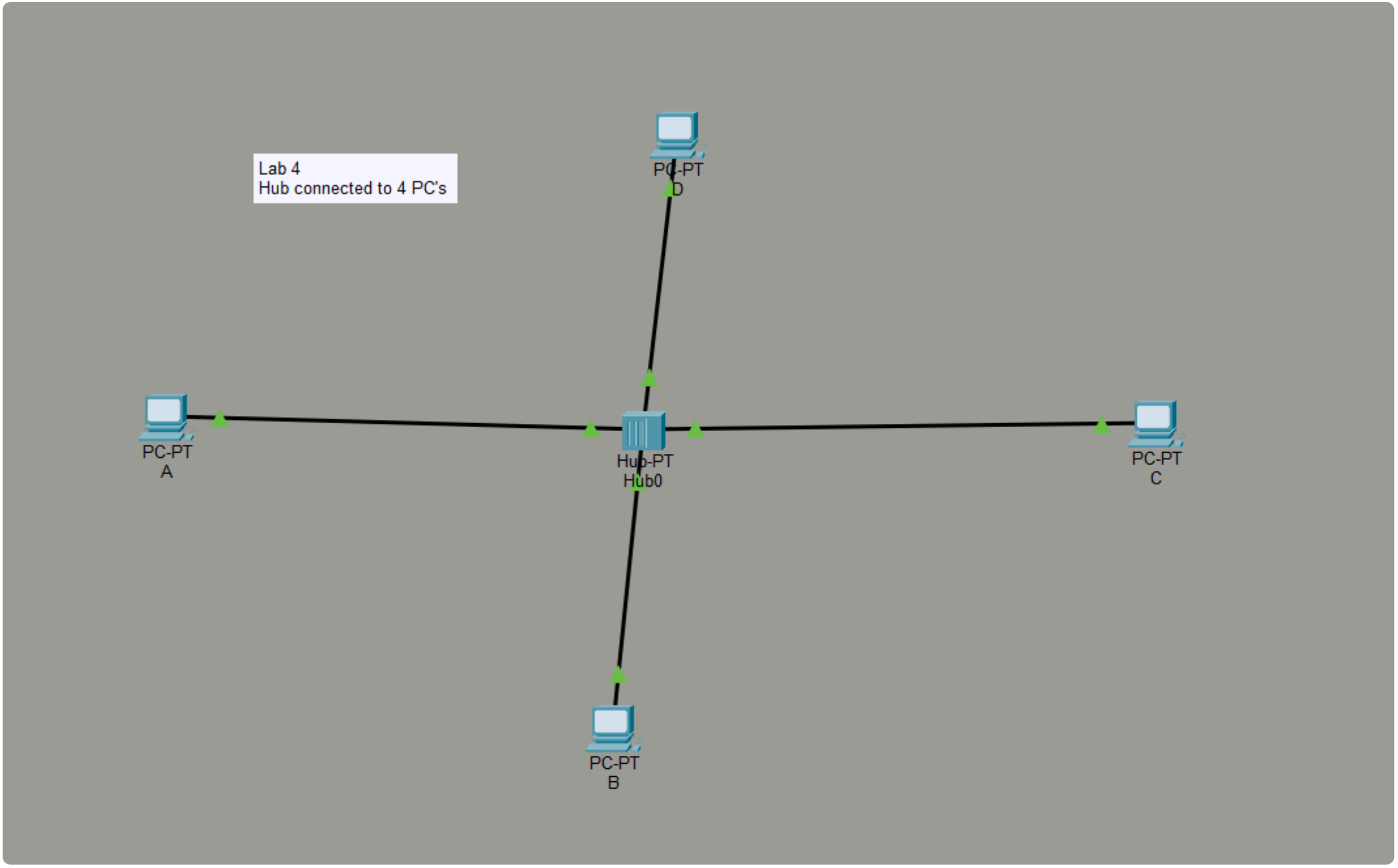
Top

Time: 00:04:51 | Realtime | Simulation

Ping Test Results

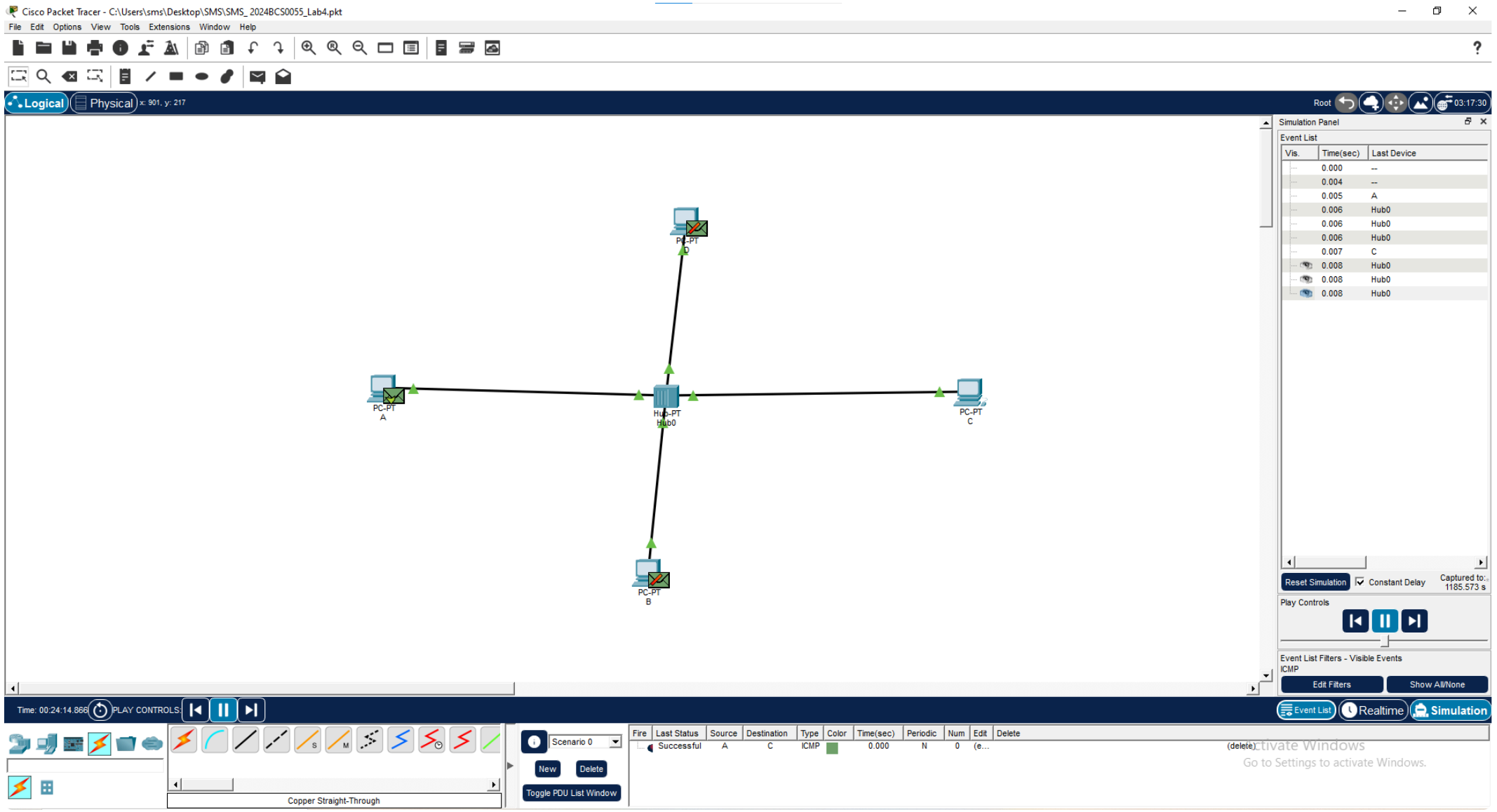


Design Screenshot



Simulation Mode

Source	Destination	Protocol	Status
PC A	PC C	ICMP	Successful



Observations

- All devices belong to the same subnet.
- Hub sends incoming frames to *ALL* connected ports.
- Every device receives the broadcasted traffic.
- Communication is shared among all devices. But depending on who the header is for, the message is accepted or received on the PC's end, but the Hub sends it to all the devices.

Inference (Hub vs Switch)

Hub

- Operates at Physical Layer (Layer 1)
- Does not store addresses
- Broadcasts frames to all ports
- No intelligence or decision-making capability

Switch

- Operates at Data Link Layer (Layer 2)
- Maintains a MAC address table
- Forwards frames based on destination MAC address
- Provides a CLI for configuration and management
- Considered an intelligent networking device

Hubs are prone to more collisions and are "dumber" whereas switches are more intelligent and perform better.

Conclusion

All PCs were successfully connected through the Hub and were able to communicate using ping. The comparison shows that a Switch is more intelligent than a Hub due to its MAC address table and configuration capabilities.